

Particle deposition and characteristics of turbulent flow in converging and diverging nozzles using Eulerian-Lagrangian approach

Md. Shahneoug Shuvo, Md Nazmus Sakib, Rezwana Rahman, Sumon Saha *

Department of Mechanical Engineering, Bangladesh University of Engineering and Technology, Dhaka, 1000, Bangladesh

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ABSTRACT

The current study conducts a thorough assessment of the deposition of aerosol particles in turbulent flows in both converging and diverging nozzles using the Lagrangian particle tracking model and the RNG $k-\epsilon$ turbulence model. The air-particle interaction depends on flow velocity, particle diameter, and types of flow sections when the particles are injected into the airflow. For the parametric simulation, Stokes number is varied from 0.0354 to 0.8265 with the corresponding particle diameter ranging from 2.00 to 10.00 μm , whereas Reynolds number is varied from 5000 to 10,000. Computational results show how the particle deposition efficiency in converging and diverging nozzles is influenced by Stokes number, diameter ratio of pipe, particle diameter, and Reynolds number. The deposition efficiency of a diverging nozzle is found to be lower than that of a converging nozzle, and the flow Reynolds number increases the deposition efficiency slightly in the diverging nozzle but gradually in the converging nozzle.

1. Introduction

The continuous adherence of particles to a surface is known as particle deposition which is due to the hydrodynamic flow conditions and favorable particle-surface interaction. Due to its numerous applications in fields such as aerosol science, gas piping networks, environmental systems, biomedicine, pharmaceuticals, ventilation, industrial systems, and others, particle deposition in a pipe has drawn more interest from researchers. Notably, particle accumulation in a duct or pipe has drawn a lot of attention since it impairs airflow and causes problems.

Air can be used in a wide range of applications and can be found in a lot of different parts of engineering. In the ventilation system, air must be filtered and conditioned with care. Cleanrooms in the semiconductor industry employ sophisticated technology to prevent the entry of any particles. Additionally, the pneumatic system relies solely on air. Air is abundant in the atmosphere; hence, industrial systems also use air in fluidics devices. Additionally, airplane cooling systems and vortex tube refrigeration are air-based.

Several factors like cross-section, surface condition, flow rate, bend configuration, etc., influence particle deposition in pipes. The diameter of the pipe can range from 1 mm (air sampling pipes) to 1 m (designed to transport fluids in industries). In addition, the pipe bend radius, Stokes

number, and Reynolds number are frequently noted as factors that influence particle deposition [1,2].

Numerous scholars have conducted numerical and experimental studies of particle deposition in various geometries and cross-sections for a wide range of particle sizes and Reynolds numbers [3–26]. In the literature, only a few experimental studies were conducted on aerosol particle deposition and dispersion in pipes. Though more costly and challenging to do, experimental research provides critical information to validate numerical studies, and the ability to derive empirical correlations that are simple to apply. For example, Montgomery and Corn [3] investigated micron particle deposition characteristics with a diameter ranging from 0.44 to 2.16 μm when subjected to a turbulent flow in a circular tube of 150 mm diameter. They showed that Reynolds number had a substantial influence on particle deposition. Saldo et al. [4] as well as Muyschondt et al. [5] examined particle deposition in ducts of varying sizes. In other studies, particle deposition characteristics in rectangular and square ducts were investigated [6–8]. Deposition in 90° bends was studied experimentally by Pui et al. [9] and McFarland et al. [10] for Reynolds numbers ranging from 10^2 to 10^4 . According to the findings of their experiment, using an exponential curve, Pui et al. [9] proposed a relationship between deposition efficiency and Stokes number. Following the work of Pui et al. [9], Brockmann [11] analyzed the influence of bend angle and Stokes number on aerosol deposition. Kim

* Corresponding author.

E-mail addresses: shahneougshuvobuet396@gmail.com (Md.S. Shuvo), nazmussakib8503@gmail.com (M.N. Sakib), rezwanar1@gmail.com (R. Rahman), sumonsaha@me.buet.ac.bd (S. Saha).

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Nomenclature			
a_1, a_2, a_3	constants for particles	$u_{p,i}$	instantaneous particle velocity (m/s)
C_c	Cunningham correction factor	y^+	dimensionless parameter for wall
C_d	drag coefficient	<i>Greek Symbols</i>	
D	hydraulic diameter (m)	η	deposition efficiency
d_p	particle diameter (m)	ν	kinematic viscosity (m^2/s)
F_d	drag force per unit mass (m/s^2)	λ	mean free path (m)
g	acceleration due to gravity (m/s^2)	τ	relaxation time of the particle (s)
k	turbulent kinetic energy (m^2/s^2)	ε	turbulent dissipation rate (m^2/s^3)
k_m	mean turbulent kinetic energy (m^2/s^2)	μ	dynamic viscosity ($\text{kg}/\text{m}\cdot\text{s}$)
N	number of particles	ρ	density (kg/m^3)
N_E	number of elements	<i>Subscripts</i>	
N_{pe}	number of particles entered	i, j	direction
N_{pt}	number of particles trapped	p	particle
$\bar{p}$	time-averaged air pressure (Pa)	pe	particles entered
Re	flow Reynolds Number	pt	particles trapped
Re_r	relative Reynolds Number	<i>Abbreviations</i>	
R_{ij}	Reynolds stress tensor ($1/\text{m}^2\text{s}^2$)	CFD	Computational Fluid Dynamics
St	Stokes number	DPM	Discrete Phase Model
t	time (s)	EWT	Enhanced Wall Treatment
u_i	instantaneous air velocity (m/s)	LES	Large Eddy Simulation
u_i'	the i^{th} component of fluctuating air velocity (m/s)	RNG	Renormalization Group
u_m	mean flow velocity (m/s)	DRW	Discrete Random Walk

et al. [12] reported experiments on particle deposition efficiency inside a bifurcation duct. Stokes number appeared to be the most critical element in particle deposition characteristics within the bifurcating airways, according to the findings of their study. Sun et al. [13] investigated particle deposition efficiency in a 90° square bend with a 10 cm side length. To assert the relationship between particle deposition and Stokes number, an empirical model and a simple assessment approach was proposed by them. Particle deposition patterns on a 180° square bend's interior curved wall were studied experimentally by Li et al. [14]. They further investigated the relationship between secondary flow, deposition pattern, and tangential velocity.

In recent times, for two-phase flows and particle deposition efficiency in bends and ducts, computational fluid dynamics (CFD) has shown its predictive capability. Hence, bends and straight ducts were studied numerically in terms of aerosol deposition [15,16]. One unifying correlation was proposed by Inthavong [17] for the particle deposition rate of micron-sized particulates for laminar flow in a 90° circular bend. He proposed a correlation for particle deposition efficiency for laminar flows by varying Stokes numbers. On the other hand, Zhang et al. [18] provided an empirical correlation and computational guidelines for Eulerian-Lagrangian particle deposition modeling in curved pipelines. Moreover, Chibbaro and Minier [19] used a Langevin probability density function model to conduct particle deposition in a turbulent pipe flow numerically. Particle deposition distribution along the axis of the circular 90° bend was estimated by Breuer et al. [20] utilizing the large-eddy simulation (LES) method. Aerosol deposition efficiency in a circular bend for turbulent flow was simulated using the LES and an effective Lagrangian tracking approach by Berrouk et al. [21]. They showed the deposition patterns on both inner and outer bends. Using the Eulerian-Lagrangian approach, Arsalanloo et al. [22] quantitatively examined particle deposition efficiency in a 90° bend of circular cross-section with swirling flows. From their study, it was discovered that larger particles having higher Stokes numbers behaved more predictably in swirling flow. On the outer bend side, Jiang et al. [23] simulated the particle deposition distribution in a 90° ventilation bend (2D). Their numerical findings agreed with the results of the experimental investigation as well as the information gathered from the earlier research. Cong et al. [24] investigated particle penetration efficiency in

turbulent and laminar flows with varied curvature ratios in a 90° circular bend. They demonstrated that decreasing the air velocity and Stokes number increased particle penetration, whereas increasing the bend curvature ratio reduced it. Besides, deposition rates of mono-dispersed particles in ducts with different contracting and expanding ratios were investigated by Lu et al. [25]. They found that particle deposition characteristics could be modified primarily by the changing of air velocity, deposition distance, and airflow structures. The simulation of Talebizadehsardari et al. [26] on particle deposition dealt with the flow of nanoparticles in an annular pipe at Reynolds number of 10,000 without considering any bend pipe/duct. They proposed a correlation that could be used to calculate the percentage of nanoparticles that were deposited in annular pipe flows.

Considering the internal resistance of particles, the Cattaneo-Christov heat flux model for heat transfer in a non-Newtonian fluid was studied by Khan and Alzahrani [27]. They found that Brownian motion and thermophoresis promoted temperature dispersion. Chu et al. [28] investigated particle deposition modeling for non-linear thermally generated flow. They demonstrated that the radiation parameter, Brownian motion, and thermophoretic constant increased the temperature of nanomaterials. Using CFD and in silico modeling, Huang et al. [29] studied the impact of aerosol deposition on the respiratory system. They demonstrated the viability of using CFD models to conduct an in-depth analysis and prediction of the aerosol deposition pattern. Brazhenko and Mochalin [30] conducted experiments and numerical simulations on a filter with a rotary cylindrical filter element. In the simulations, it was found that the velocity in the pores of the perforated filter element increased regionally. Roberts et al. [31] conducted a validation and analysis of a CFD-DPM approach for modeling dust particles. Results revealed that CFD-DPM could be employed practically in the design of cleaning and disinfection systems. Wang et al. [32] examined elliptical particle deposition by a numerical investigation. They showed that the particle's density ratio was proportional to its time-averaged deposition velocity. Lin et al. [33] conducted research utilizing Monte Carlo simulation to investigate the process of slag particle deposition. They looked into how temperature affected the deposition process. Recently, a review of the influence of air ions on the indoor environment was done by Hartmann and Kriegl [34]. They

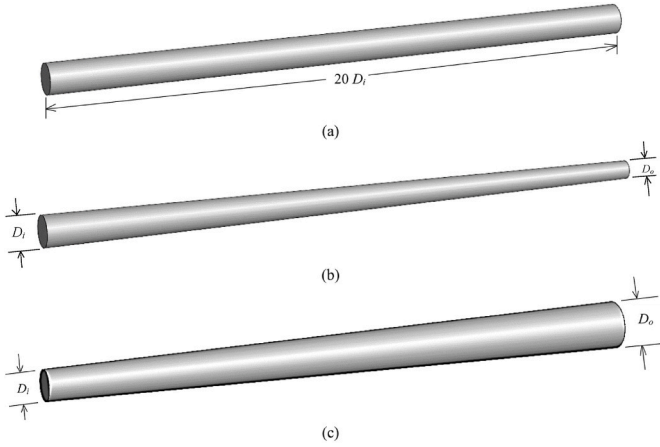


Fig. 1. Three-dimensional sketch of the computational domains considered in the present study: (a) straight pipe ($\lambda = 1.00$), (b) converging nozzle ($\lambda = 1.67$), and (c) diverging nozzle ($\lambda = 1.60$).

concluded that air ions tended to reduce the deposition of particles in the air. Besides, Alqallaf and Teixeira [35] investigated the impact of solid particle erosion on engine performance using a numerical model.

From the above literature review, it can be inferred that many studies have focused on particle deposition in circular and square cross-sectional pipes and ducts with straight and 90° bends. However, the expansion and contraction of pipe cross-sections, which are well-known to be utilized for many applications, have not yet been explored. Industrial systems make considerable use of converging and diverging nozzles in medical products, bioengineering, ventilation, control equipment, etc. Hence, particle deposition in pipes with converging and diverging nozzles is the primary attention of this study. The present concern is to explore how the deposition efficiency changes based on diameter ratios of pipe, particle diameters, Stokes number, and Reynolds number.

2. Physical modelling

In this study, one converging nozzle, one diverging nozzle, and a straight pipe are used to model the airflow carrying aerosol particles. Fig. 1 depicts the three-dimensional sketch of those geometries. The diameter ratio ($\lambda = D_l/D_s$, where D_l is the larger diameter and D_s is the shorter diameter of the nozzle) is 1.0 for the straight pipe, and more than 1.0 for converging and diverging nozzles. The inlet diameter of the pipe as well as the nozzles is $D_i = 10$ mm. The length of the pipe and the nozzles are taken as $20D_i$.

3. Mathematical modelling

3.1. Modeling of continuous phase

Using Reynolds-Averaged Navier-Stokes, the turbulent flow of air in an industrial pipeline including a converging or a diverging nozzle with the above-mentioned geometrical size is studied. The RANS equations for a statistically steady-state, incompressible, continuous, turbulent air flow are given as follows [36]:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \frac{\mu}{\rho} \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j} R_{ij}, \quad (2)$$

where $\bar{u}_i$ denotes the time-averaged air velocity, $\bar{p}$ denotes the time-averaged air pressure, x_i denotes the position in space, $R_{ij} =$

$\overline{u'_i u'_j}$ denotes the Reynolds stress tensor, $u'_i = u_i - \bar{u}_i$ denotes the i th component of fluctuating air velocity, and u_i denotes the instantaneous air velocity. To solve the continuous phase, Eulerian approach is used. The dynamic viscosity and the density of air considered in this study are $\mu = 1.7894 \times 10^{-5}$ kg/ms and $\rho = 1.225$ kg/m³ respectively.

The RNG k - ϵ turbulence model improves the accuracy for swirling flows. Additionally, compared to the traditional k - ϵ turbulence model, it is more accurate for a broader category of flows. The RNG k - ϵ model also requires less computing time than the RSM model [37]. Hence, to calculate turbulent parameters, e.g., turbulent kinetic energy (k) and turbulent dissipation rate (ϵ), the RNG k - ϵ turbulent model is employed referencing [38]:

$$\frac{\partial}{\partial x_i} (\rho k \bar{u}_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j} \right) + G_k + G_b - \rho \epsilon - Y_M + S_k, \quad (3)$$

$$\frac{\partial}{\partial x_i} (\rho \epsilon \bar{u}_i) = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{eff} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon + S_\epsilon. \quad (4)$$

In these equations, S_k and S_ϵ are user-defined source terms, G_k denotes turbulent kinetic energy development owing to the mean velocity gradients, G_b denotes turbulent kinetic energy development because of density variation, α_k and α_ϵ denote the effective inverse Prandtl numbers for k and ϵ respectively, Y_M denotes the contribution of compressible turbulence's variable dilatation to the overall dissipation rate. In RNG theory, the scale elimination procedure yields a differential equation for turbulent viscosity (μ_{eff}) according to Ref. [38]:

$$d \left(\frac{\rho^2 k}{\sqrt{\epsilon \mu}} \right) = 1.72 \frac{\hat{v}}{\sqrt{(\hat{v})^3 - 1 + C_v}} d\hat{v}, \quad (5)$$

where,

$$\hat{v} = \frac{\mu_{eff}}{\mu}, \mu_{eff} = \rho C_\mu \frac{k^2}{\epsilon}. \quad (6)$$

R_ϵ and $C_{3\epsilon}$ in (4) are defined as,

$$R_\epsilon = \frac{C_\mu \rho \zeta^3 (1 - \zeta/\zeta_0)}{1 + \beta \zeta^3} \frac{\epsilon^2}{k}, \quad (7)$$

where,

$$\zeta = \frac{S_k}{\epsilon}, C_{3\epsilon} = \tanh \left| \frac{\bar{u}_y}{\bar{u}_x} \right|, \quad (8)$$

where, $\bar{u}_x$ and $\bar{u}_y$ are the flow velocity components perpendicular and parallel to the gravitational vector. The constants for RNG k - ϵ model are as follows [38]:

$$C_v = 100, C_{1\epsilon} = 1.42, C_{2\epsilon} = 1.68, C_\mu = 0.0845, \zeta_0 = 4.38, \beta = 0.012. \quad (9)$$

3.2. Modeling of discrete phase

The particles are thought to be smooth, spherical, traveling in a Lagrangian reference frame, and dilute (less than 10% of volume fraction) so that their movements do not affect the airflow. For this reason, the particles' paths are tracked using one-way coupling. The discrete phase model (DPM) is used to trace the movement of individual particles as they pass through the flow field. The equation of motion of the particle [19] as suggested below is solved using the Runge-Kutta discretization method [39]:

$$\frac{du_{p,i}}{dt} = F_d + \frac{g(\rho_p - \rho)}{\rho_p}, \quad (10)$$

where ρ_p is the mass density of the particle, g is the gravitational acceleration, $u_{p,i}$ is the instantaneous particle velocity, and F_d indicates the

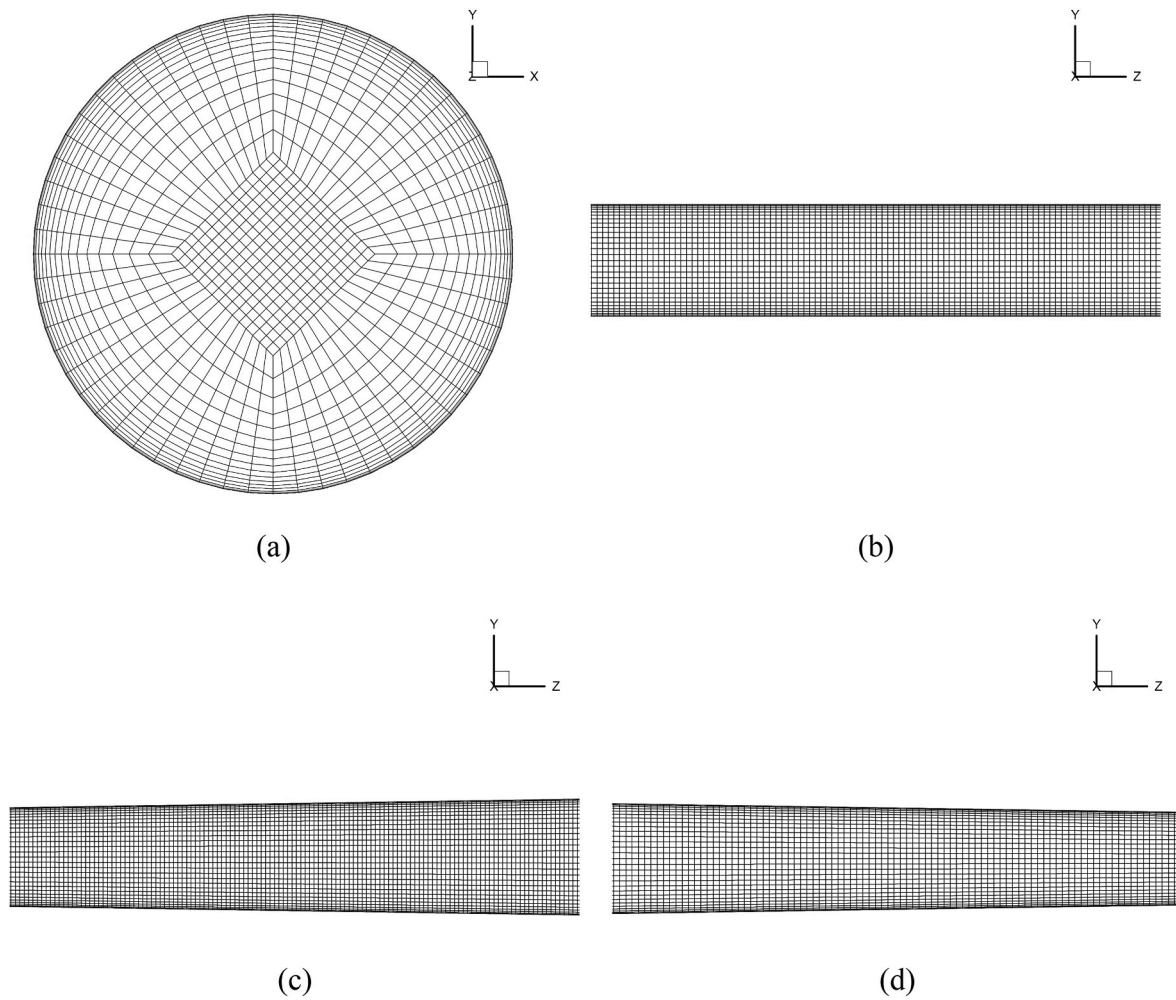


Fig. 2. Distribution of O-type structured mesh of the pipe: (a) cross-section, (b) straight section, (c) diverging section, and (d) converging section.

drag force per unit flow of mass which is further defined as follows [19]:

$$F_d = \frac{18\mu}{\rho_p d_p^2} \frac{C_d Re_r}{24} (u_i - u_{p,i}) = \frac{(u_i - u_{p,i})}{\tau_p}, \quad (11)$$

where d_p is the particle diameter, τ_p is the relaxation time of the particle, and C_d is the drag coefficient of the particle. In spherical drag law, the definition of drag coefficient is given by Morsi and Alexandar [40] as:

$$C_d = a_1 + \frac{a_2}{Re_r} + \frac{a_3}{Re_r^2}, \quad (12)$$

where a_1, a_2, a_3 are constants over various ranges of particles of relative Reynolds number (Re_r) which is calculated as follows:

$$Re_r = \frac{\rho d_p |u_{p,i} - u_i|}{\mu}, \quad (13)$$

where $|u_{p,i} - u_i|$ denotes the magnitude of the relative velocity. Similarly, the flow Reynolds number is indicated as,

$$Re = \frac{\rho u_m D}{\mu}, \quad (14)$$

where u_m is the mean flow velocity.

During the turbulent flow of the fluid, particle trajectories are influenced by fluctuating velocity components. To address this in the calculation (10), discretized random walk (DRW) model governed by stochastic particle tracking is introduced. Moreover, additional forces in

(10) can also be considered depending on the flow condition. For example, the Brownian force and the Staffman's lift force become significant for particles below micron level [19]. However, these forces can be neglected as the particle diameter considered in this study varies from $d_p = 2.00$ – $10.00 \mu\text{m}$ with a density of $\rho_p = 895 \text{ kg/m}^3$. The entire study is free of any temperature gradient and the density ratio of fluid to the particle is quite negligible. Hence, the effects of the thermophoretic force, virtual mass force, and pressure gradients are also neglected. In this analysis, any Magnus force in (10) has not been considered. This is because the Magnus force is negligible in comparison to the drag force when the particle size or spin velocity is small [41].

3.3. Turbulent dispersion

The interactions between turbulent eddies and particles are evaluated using the DRW model. During the lifetime of the eddy, it is approximated that the instantaneous fluctuating components of airflow follow the Gaussian probability distribution given as [38]:

$$u'_x = \xi_1 \sqrt{u'_x u'_x}, u'_y = \xi_2 \sqrt{u'_y u'_y}, u'_z = \xi_3 \sqrt{u'_z u'_z}, \quad (15)$$

where, ξ_1, ξ_2, ξ_3 , are random numbers of the normal distribution with unit variance and zero mean. To utilize the above approximation, the root-mean-square value of fluctuation in (15) is considered to be isotropic and equal to $\sqrt{(2k/3)}$. Each eddy's characteristic time scale τ_e is given as follows [38]:

$$\tau_e = 2T_L, \quad (16)$$

where T_L denotes the Lagrangian integral time, which is roughly equivalent to:

$$T_L \approx \frac{0.15k}{\varepsilon}. \quad (17)$$

3.4. Governing parameters and boundary conditions

The particle deposition efficiency is governed by the Stokes number, which is defined as the relationship between the relaxation time (τ_p) of a particle and the response timing (t_s) of the system. A particle with a high Stokes number is dictated by its momentum and persists along its initial route, but a particle with a lower Stokes number maintains fluid streamlines. In the present study, several injections are created based on Stokes number which is expressed according to Pui et al. [9] as follows:

$$St = \frac{\tau_p}{t_s} = \frac{C_c \rho_p d_p^2 / 18\mu}{(D/2)/u_m} = \frac{C_c \rho_p d_p^2 u_m}{9\mu D}, \quad (18)$$

where C_c denotes the Cunningham correction factor which is defined as follows:

$$C_c = 1 + \frac{\sigma}{d_p} (2.514 + 0.8e^{-0.55d_p/\sigma}), \quad (19)$$

where σ is the air molecule's mean free path.

The experiments conducted by Pui et al. [9] suggested that the ricochet of the particles should not be considered. Therefore, “trap” (calculations of the particle's route are concluded as soon as it touches the wall) is used as the DPM boundary condition of the wall, while the pressure outlet is employed for the outlet, and the fully developed profiles (velocity, turbulent kinetic energy, and specific dissipation energy) are used for the inlet. For simulating a fully developed airflow, at first, the turbulent airflow is modeled by simulating a single-phase steady-state flow in a 3D long straight pipe [42]. Then, at the inlet of the 3D computational geometry, results (velocity, turbulent kinetic energy, and specific dissipation energy) obtained from the first study are coupled. The particles are injected 10 mm distance away from the inlet. It guarantees that no particles are left behind and that they are evenly dispersed throughout the flow. Particle size varies from injection to injection. Each injection contains spherical-shaped particles of the same size with a release velocity that is equal to the mean velocity of the continuous phase. Turbulence parameters affect the deposition efficiency in the regime close to the wall. In the near-wall zone, where momentum, as well as other scalar transports, are strongest, the resolution variables have significant gradients. Thus, the near-wall accuracy of the model determines how well it predicts turbulent flow that is restricted by the walls. The viscosity-influenced zone can be resolved with finer meshes all the way to the wall using enhanced wall treatment (EWT) and the turbulence near the wall can be appropriately captured. Hence, EWT option is considered to improve the accuracy of the current numerical solution.

To measure the performance of the proposed system, particle deposition efficiency η is calculated as follows [17]:

$$\eta = \frac{N_{pt}}{N_{pe}}, \quad (20)$$

where N_{pe} is the number of particles entered in the system, and N_{pt} is the number of particles trapped.

4. Numerical simulation and validation

The computational domain is discretized through a finite volume technique, and the corresponding governing equations are solved using the commercial CFD package ‘ANSYS CFD v14.5’. The SIMPLE method is

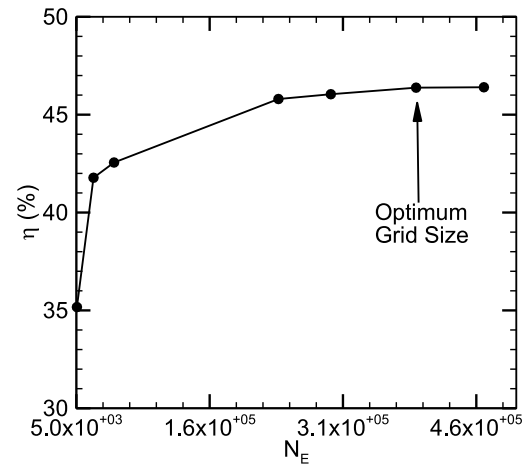


Fig. 3. Grid independent check for straight pipe at $St = 0.8265$ and $Re = 10,000$.

used to combine the velocity and the pressure fields. A discretization of the velocity, pressure, turbulent kinetic energy, and dissipation rate is achieved using the second-order upwind method. Absolute tolerance of 10^{-5} for setting stopping criteria is selected to monitor the convergence of the solution.

The meshing used within the computational domain is depicted in Fig. 2. For this study, an O-type structured mesh is utilized, and refinement has applied near the wall while a singularity at the center of the nozzle is avoided. The multi-zone meshing approach is used for the fluid domain with uniform surface meshing. For near wall refinement in the center of the mesh, the growth factor for the cells is 1.2, whereas the first cell near the wall is 0.015 mm thick. A grid-independent test is used to find the best mesh layout. Eight alternative mesh configurations are evaluated, and the estimated deposition efficiency for a Stokes number of 0.8265 is shown in Fig. 3, which reveals that a mesh structure with the number of elements $N_E = 3,87,260$ along with elements size of 0.50 mm enables a grid-independent solution to the existing study. Further increment of the number of nodes and elements has no significant change on deposition efficiency. For the selected meshing, the value of $y^+ = \rho u_\tau y / \mu$, where u_τ is the friction velocity, is found to be around 1.4, which is acceptable according to the ANSYS FLUENT user's guide [43]. Hence the application of EWT in this model is justified.

For the validation of the present turbulence model, variation of axial mean velocity ($\bar{u}_z$) with the experimental data of Eggels et al. [44] at $Re = 10,000$ is plotted in Fig. 4(a). The test segment in their experiment consisted of a bell-mouth intake followed by three identically honed sections of aluminum pipe with $D_i = 95.4$ mm with a total length of $112.5D_i$. The RNG $k-\varepsilon$ turbulence model with EWT is chosen for this investigation. It is observed that experimental data is in good agreement near the wall and at the center of the pipe, but slightly deviates in between them.

Verification of the present deposition model is carried out based on the experimental findings and fitting curve published by Pui et al. [9] as well as the numerical results obtained by Zhang et al. [18]. Pui et al. [9] examined monodispersed liquid aerosol deposition efficiency having $\rho_p = 895$ kg/m³ in a 90° circular bend pipe with a curvature ratio of 5.60 and a diameter of 8.51 mm at $Re = 10,000$. Because of using the liquid aerosol, the particles cling to the near wall and do not bounce when they come into touch with the bend's walls. Hence, for DPM, the “trap” boundary condition is applied to the bend's walls. The simulation settings utilized in the verification and validation of the present deposition model are equivalent to the conditions employed in the experiment conducted by Pui et al. [9]. Fig. 4(b) shows the variation in computational and experimental findings of Pui et al. [9] in terms of deposition efficiency against Stokes number. The present simulation results closely

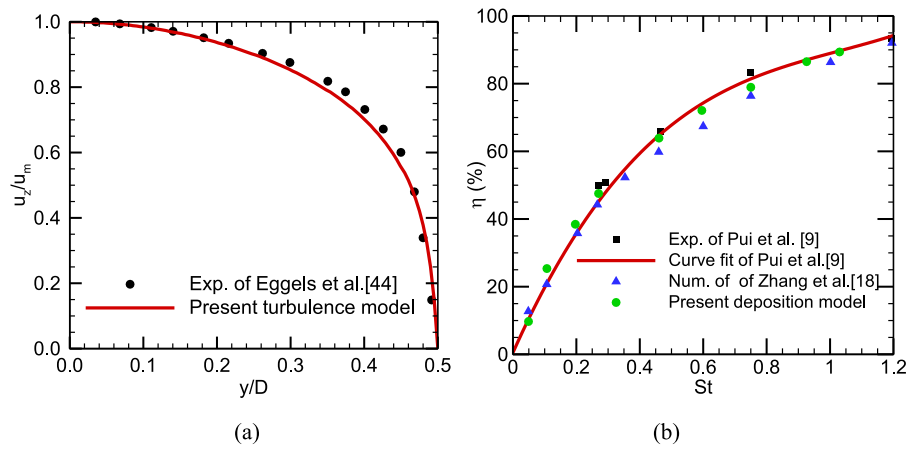


Fig. 4. (a) Resemblance of the current turbulence model with the experimental data of Eggels et al. [44], and (b) the current deposition model in terms of particle deposition efficiency against the data available in previous literature (color online).

Table 1

Selection of geometric and governing parameters for the current investigation.

Parameter	Values of interest
Pipe inlet diameter, D_i mm	10
Pipe length, L mm	200
Diameter ratio, λ	1.00, 1.25, 1.40, 1.60, 1.67
Particle diameter, d_p μm	2.00, 3.00, 4.10, 4.82, 6.32, 7.20, 8.09, 9.00, 9.50, 10.00
Reynolds number, Re	5000, 6000, 7000, 8000, 9000, 10,000

match the experimental findings indicating the accuracy and reliability of the computational approach and the deposition model used in this study.

5. Results and discussion

To investigate particle deposition and fluid flow characteristics of aerosol particles, three distinct diameter ratios for converging nozzles ($\lambda = 1.67, 1.40, 1.25$) and diverging nozzles ($\lambda = 1.25, 1.40, 1.60$) with straight pipe ($\lambda = 1.00$) are selected. For maintaining turbulent flow inside the test section, Reynolds numbers considered in this investigation vary from 5000 to 10,000. Table 1 lists the selected geometric and

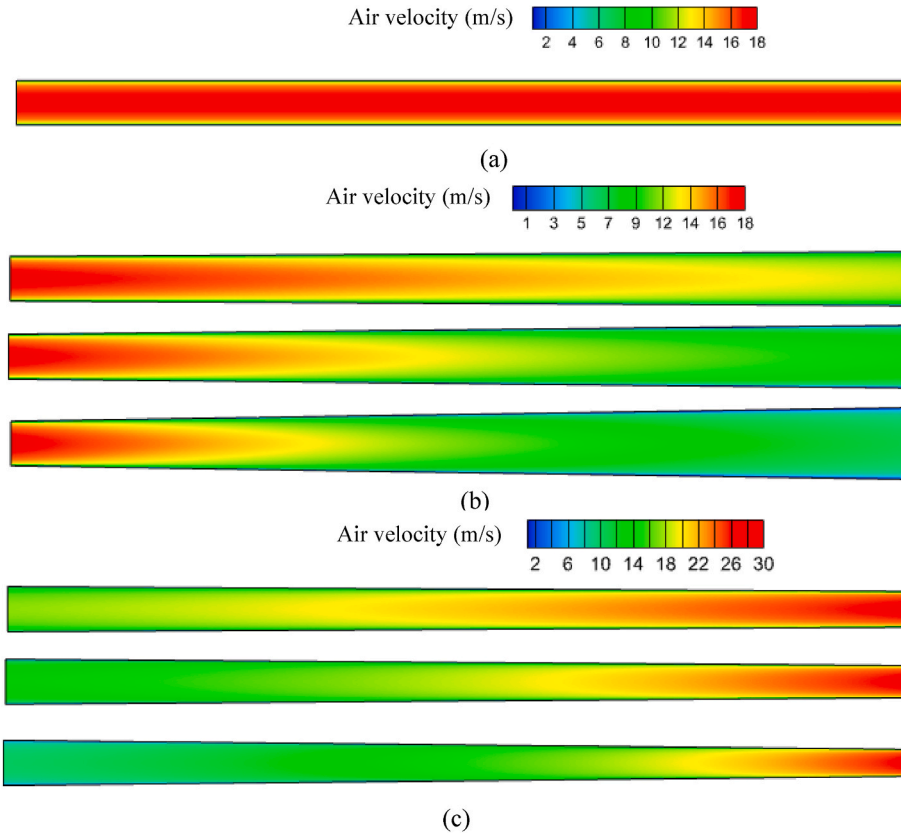


Fig. 5. Visualization of velocity fields inside (a) straight pipe ($\lambda = 1.00$), (b) diverging nozzle ($\lambda = 1.25, 1.40, 1.60$ from top to bottom), and (c) converging nozzle ($\lambda = 1.25, 1.40, 1.67$ from top to bottom) at $Re = 10,000$ (color online).

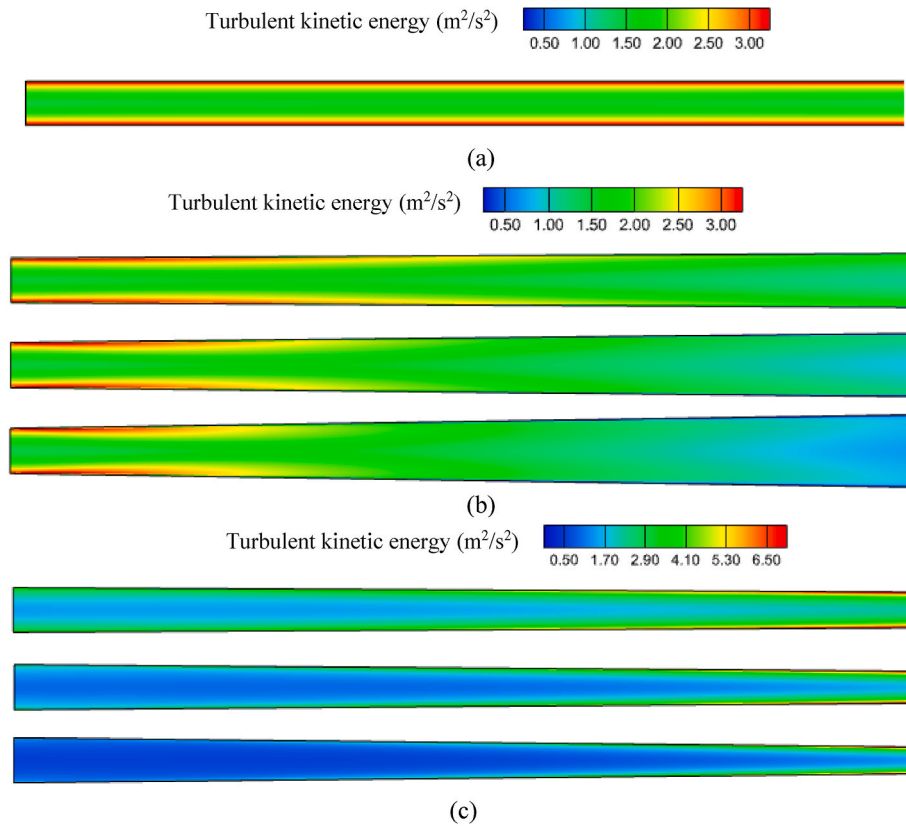


Fig. 6. Visualization of turbulent kinetic energy fields inside (a) straight pipe ($\lambda = 1.00$), (b) diverging nozzle ($\lambda = 1.25, 1.40, 1.60$ from top to bottom), and (c) converging nozzle ($\lambda = 1.25, 1.40, 1.67$ from top to bottom) at $Re = 10,000$ (color online).

governing parameters used in the current investigation.

5.1. Visualization of turbulent airflow characteristics

Fig. 5 depicts the velocity fields of air flowing through the straight pipe, diverging nozzle, and converging nozzle at $Re = 10,000$. As illustrated in Fig. 5(a), the flow velocity of air in a straight pipe varies only near the boundary wall, whereas it remains constant at the center of the pipe. On the other hand, flow fields of air in diverging and converging nozzles are more complex than the straight ones. According to Fig. 5(b), the air velocity decreases in diverging nozzles along the streamwise direction due to the greater downstream flow area. In the case of converging nozzles, large changes in air velocity can be observed throughout the flow fields (see Fig. 5(c)). The velocity of air for a converging nozzle is increased in the flow direction due to the smaller downstream area. These tendencies are accentuated when the diameter ratios of the diverging as well as the converging nozzles are increased.

Next, turbulent kinetic energy fields inside the straight pipe, diverging nozzle, and converging nozzle at $Re = 10,000$ are visualized in Fig. 6. In contrast to the velocity fields, where the maximum value is observed at the center, turbulent kinetic energy is maximum near the walls due to fluctuating velocity and progressively decreases to its minimum as it approaches the center. Despite this, distributions of turbulent kinetic energy in diverging and converging nozzles are more complex than in straight pipes. The converging nozzles possess large changes in turbulent kinetic energy throughout the flow fields (see Fig. 6 (c)). This is because passing through the nozzle intensifies airflow turbulence and the impact of eddies. Fig. 6 depicts these changes. The complicated turbulent kinetic energy fields may lead to more complicated particle deposition patterns.

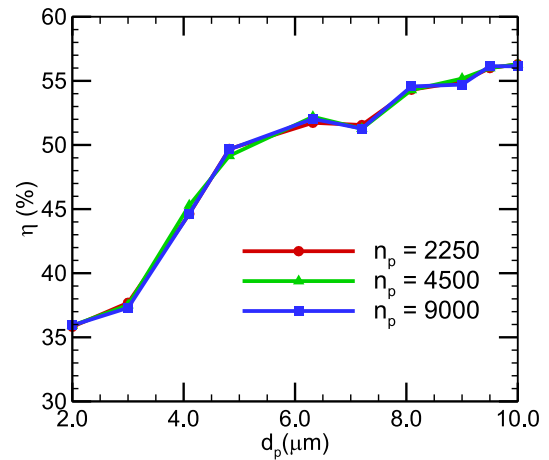


Fig. 7. Impact of the number of injected particles on aerosol deposition efficiency as a function of particle diameters for the converging nozzle ($\lambda = 1.67$) at $Re = 10,000$ (color online).

5.2. Visualization of aerosol particle deposition

Following the same methodology as the validated models [9,18], visualization of aerosol particle deposition in converging and diverging nozzles has been carried out. A CFD simulation is conducted for three different numbers of injected particles (2250, 4500, and 9000 particles respectively) to explore the effect of the number of injected particles on the deposition of aerosol particles in the converging nozzle. Particle deposition is not significantly affected by the increment of the number of injected particles as illustrated in Fig. 7. Therefore, in general, it can be

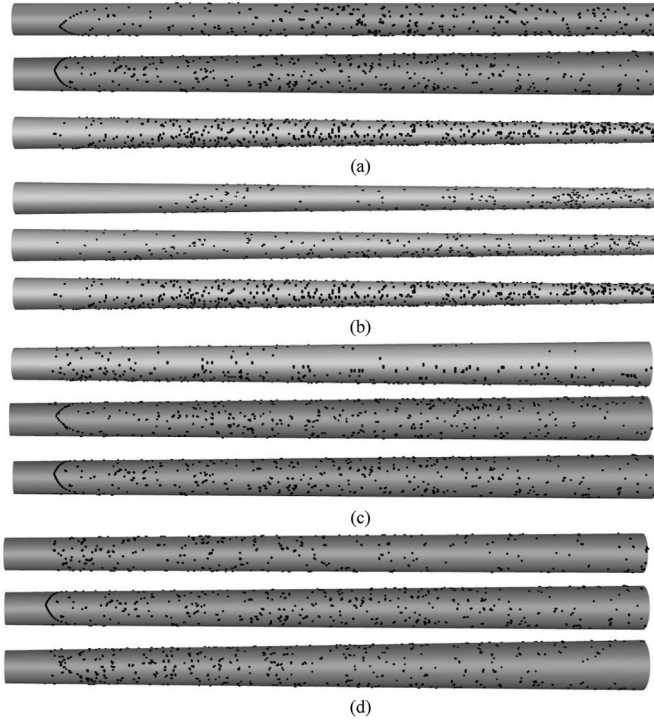


Fig. 8. (a) Deposited particles in straight pipe ($\lambda = 1.00$), diverging nozzle ($\lambda = 1.40$), converging nozzle ($\lambda = 1.67$) respectively from top at $Re = 10,000$, (b) deposited particles in converging nozzle ($\lambda = 1.67$) for $d_p = 2, 4.82$, and $9 \mu m$ respectively from top at $Re = 10,000$, (c) deposited particles in diverging nozzle ($\lambda = 1.40$) for $Re = 5000, 7000, 10,000$ respectively from top at $d_p = 9 \mu m$, and (d) deposited particles in diverging nozzle for $\lambda = 1.25, 1.40, 1.60$ respectively from top at $Re = 10,000$ and $d_p = 9 \mu m$.

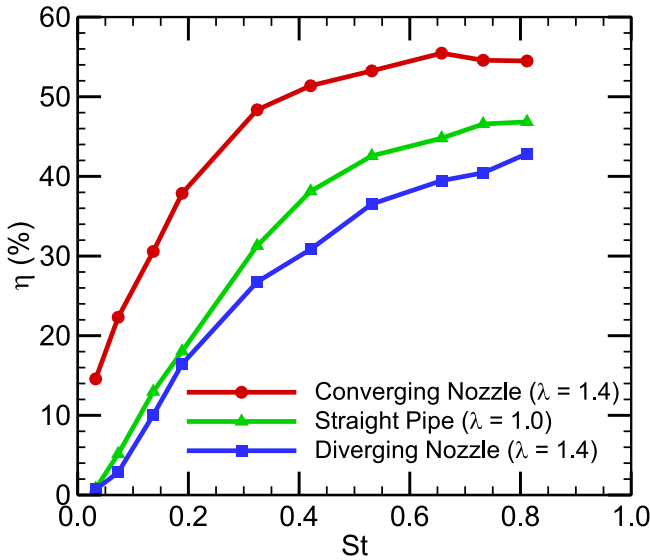


Fig. 9. Effect of Stokes number on deposition efficiency for converging nozzle ($\lambda = 1.4$), diverging nozzle ($\lambda = 1.4$), and straight pipe ($\lambda = 1.0$) at $Re = 10,000$.

said that aerosol particle deposition is unaffected by particle concentration like the outcomes of prior research [45]. For further analysis, 9000 particles are injected in the present investigation.

Deposition characteristics may be attributed to the pipe geometry, turbulent flow structures, large particles inertia, etc., all of which influence the number of particles deposited. Fig. 8 presents the

visualization of the deposited particles for straight pipe, converging nozzle, and diverging nozzle for different combinations of governing and geometric parameters.

5.3. Effect of Stokes number on deposition efficiency

Fig. 9 illustrates the impact of Stokes number on deposition efficiency for converging nozzle ($\lambda = 1.4$), diverging nozzle ($\lambda = 1.4$), and straight pipe ($\lambda = 1.0$) at $Re = 10,000$. For smaller Stokes numbers, the deposition efficiency on each configuration is insignificant since the particles with lower inertia continue to follow the airflow closely. On the other hand, as the Stokes number increases, the larger particles' response time decreases; hence, they cannot flow along the fluid flow entirely and are finally deposited on the wall, which results in higher deposition efficiency. However, this increment is much more pronounced in the converging nozzle than in the other two. Specifically, this is due to an increase in turbulent kinetic energy (TKE) near the wall along the flow direction in the converging nozzle. The high TKE would aid the turbulent dispersion of the particles. Fig. 6 also depicts that the straight pipe's TKE along the flow is higher than that of the diverging nozzle. As a result, the straight pipe's deposition efficiency is generally higher than the diverging nozzle's for a given Stokes number.

5.4. Effect of diameter ratio on deposition efficiency with Reynolds number

To investigate the impact of diameter ratios on aerosol particle deposition, a plot of deposition efficiency versus Reynolds numbers for straight pipe ($\lambda = 1$) and converging nozzles ($\lambda = 1.25, 1.40, 1.67$) at $d_p = 7.2 \mu m$ is displayed in Fig. 10(a). The efficiency of particle deposition in a converging nozzle is greater than that of a straight pipe. As the diameter ratio rises, the rate of particle deposition almost doubles for higher Reynolds numbers. This is due to the blockage in the flow caused by the converging segment, a subsequent reduction in deposition distance, and a rise in turbulent kinetic energy, and in the case of a larger diameter ratio, the value of turbulent kinetic energy near the wall is boosted (see Fig. 6). The turbulence of the particles would be aided by the high turbulent kinetic energy and thus results in the higher deposition. Again, to explore the same impact of diameter ratios for straight pipe ($\lambda = 1.00$) along with diverging nozzles ($\lambda = 1.25, 1.40$, and 1.60), another plot is displayed in Fig. 10(b). For a diverging nozzle, deposition is lower than that of a straight pipe ($\lambda = 1$). As the diameter ratios increase, the effectiveness of deposition diminishes. This is because when the diameter ratio increases for divergent nozzles, the deposition distance enhances, and the air velocity and TKE decrease along the flow. The impact of diameter ratio is very dominant in the converging nozzle than in the diverging nozzle. Additionally, due to the high inertia of the particles, the deposition efficiency is found to be higher for high Reynolds numbers in both nozzles.

5.5. Effect of particle diameter on deposition efficiency with Reynolds number

Numerical estimation of the particle deposition is conducted by plotting deposition efficiency versus Reynolds numbers for four different particle diameters ($d_p = 2.00, 4.82, 7.2, 9.00 \mu m$) in converging and diverging ($\lambda = 1.25$) nozzles as shown in Fig. 11 (a) and (b) respectively. The deposition efficiency of aerosol particles raises as the air velocity increases for both nozzles. This is due to the increasing strength of the turbulence. Turbulence eddies capture many smaller diameter particles, which are then transported to deposit on the wall. On the other hand, mass inertia is the driving factor for the deposition of large particles. As a result, their deposition efficiency increases almost linearly as the Reynolds number increases. Furthermore, particle deposition in the converging nozzle grows faster with increasing Re than that in the diverging nozzle. For both nozzles, the deposition efficiency increases

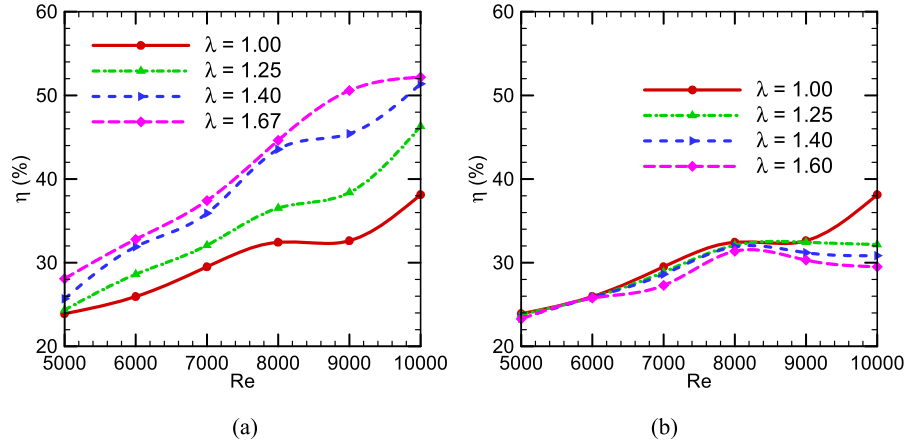


Fig. 10. Effect of diameter ratios on deposition efficiency as a function of Reynolds numbers for (a) $\lambda = 1$ (straight pipe) and 1.25, 1.40, 1.67 (converging nozzles), (b) $\lambda = 1$ (straight pipe) and 1.25, 1.40, 1.60 (diverging nozzles) at $d_p = 7.2 \mu\text{m}$ (color online).

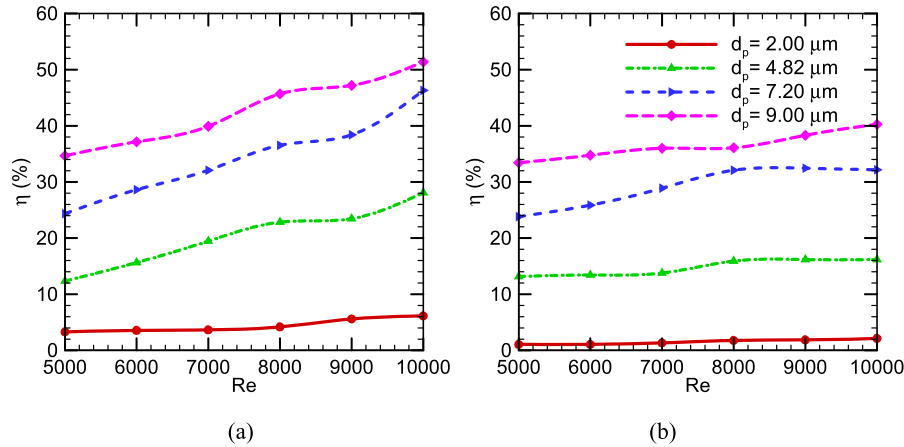


Fig. 11. Variation of deposition efficiency with Reynolds number for different particle diameters in (a) converging and (b) diverging nozzles ($\lambda = 1.25$) (color online).

with rising particle diameter. This could be owing to the high inertia of the large particles, which causes them to be less able to follow the direction of the airflow. Moreover, the gravitational pull has a considerable impact on these particles causing them to deposit on the walls. On the other hand, smaller particles due to less inertia are bound to follow the path of the airflow and have less tendency to deposit.

6. Conclusion

In this study, the aerosol particle deposition characteristics are investigated in straight pipe, converging, and diverging nozzles with fully developed turbulent flow at the nozzle's inlet using the Eulerian-Lagrangian discrete phase model after the RNG $k-\epsilon$ turbulence model. Particle sizes in the range of 2.00–10.00 μm are considered, and the parameters impacting particle deposition are computationally investigated using the finite volume method based CFD program ANSYS FLUENT. The simulation method predicts the deposition efficiency of experimental work with high accuracy. The impacts of air velocity and diameter ratio on particle deposition onto the nozzle surface are investigated. It is possible to reach the following conclusions:

- Particle deposition efficiency is unaffected by the number of particles entering the computational domain, but it increases with the Stokes number.
- The main reasons for deposition lowering for diverging nozzles relative to straight pipes are an increase in deposition distance and a slowdown of air velocity. On the other hand, with a converging nozzle, particle deposition is increased by a reduction in the deposition distance, an increase in air velocity, and the blocking action of the walls.
- The deposition efficiency of a diverging nozzle is lower than that of a converging nozzle when the inlet diameter of both nozzles is identical. As the diameter ratio of the diverging nozzle grows, the efficiency of deposition reduces. However, the efficiency of deposition in a converging nozzle increases as the diameter ratio increases. This is because, with the increment of diameter ratio, blocking action increases for the converging nozzle, whereas the diverging nozzle allows for further reduction in velocity.
- The results show that larger particles deposit more effectively than smaller ones, which is associated with the larger particles' higher inertia. As a result, small particles may readily follow streamlines and not accumulate on the wall surface.
- Reynolds number has a significant impact on deposition. Deposition efficiency increases as the Reynolds number increases for all configurations, but it is more dominant for the converging nozzle. This can be attributed to the greater increase in velocity and turbulent kinetic energy of the converging nozzle for a given Reynolds number than in the other two configurations.

- The results demonstrate that raising Reynolds number enhances the deposition efficiency slightly in diverging nozzles, but gradually raises the deposition efficiency in converging nozzles.

Credit author statement

Md. Shahneoug Shuvo: Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Md Nazmus Sakib:** Data curation, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Rezwana Rahman:** Formal analysis, Visualization, Writing – review & editing. **Sumon Saha:** Conceptualization, Supervision, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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